Two-Tailed Confidence Intervals and Their Interpretation

1. What Does "Two-Tailed" Mean?

A **two-tailed** confidence interval for a probability (or any parameter) means that the uncertainty in the estimate is accounted for on *both* the lower and upper sides. Specifically, for a 95% confidence interval:

$$\alpha = 1 - 0.95 = 0.05,$$

and this α is typically split equally into two tails of the distribution. Thus, each tail excludes:

$$\frac{\alpha}{2} = 0.025 \quad \text{(or 2.5\%)}.$$

In practical terms, a two-tailed interval implies that:

- There is a 2.5% chance (in a frequentist sense) that the true probability is *above* the upper limit of the confidence interval.
- There is a 2.5% chance that the true probability is *below* the lower limit.

Hence, the 95% confidence interval contains the true parameter in the "middle" 95% of the distribution.

2. Why Is Each Probability Presented as a Range?

When you estimate a probability (e.g., $P(2 \mid 1)$), you only observe this probability empirically from a certain number of trials. Because of finite sample sizes and natural variability, you do not know the *exact* true probability. Confidence intervals provide a range of values that are *plausible* for the true probability, given the observed data.

For instance:

$$P(2 \mid 1) = 0.6285 \quad \text{(observed)},$$

and its 95% CI might be:

$$[0.6012, 0.6559].$$

This means that, based on the sample data:

- We estimate the probability of going from 1 pellet to 2 pellets to be around 0.6285.
- We are 95% confident (in the frequentist sense) that the true probability lies between 0.6012 and 0.6559.

The range reflects the **uncertainty** in the estimate. If you had infinitely many observations of $(1 \to 2)$, the observed probability would converge to the true probability and the confidence interval would *shrink* (i.e., become very narrow).

3. Why Wilson Score Intervals in Particular?

The Wilson score interval is often preferred over simpler methods (e.g., the "Wald" or normal-approximation interval) when dealing with probabilities, because:

- It generally provides better coverage properties, especially when $\hat{p}$ is near 0 or 1.
- It avoids some biases introduced by approximating with a normal distribution at the edges.

Summary

- **Two-tailed intervals** split the total α error in half, one portion in each tail, ensuring that both possible underestimation and overestimation of the true parameter are accounted for.
- **Confidence intervals** are expressed as a range because the true probability is not known with certainty, and the interval quantifies the plausible values that the true probability might take, based on your data.
- Using **Wilson score intervals** often produces more accurate and stable confidence intervals for proportions or probabilities, particularly when sample sizes are small or the probability is near 0 or 1.